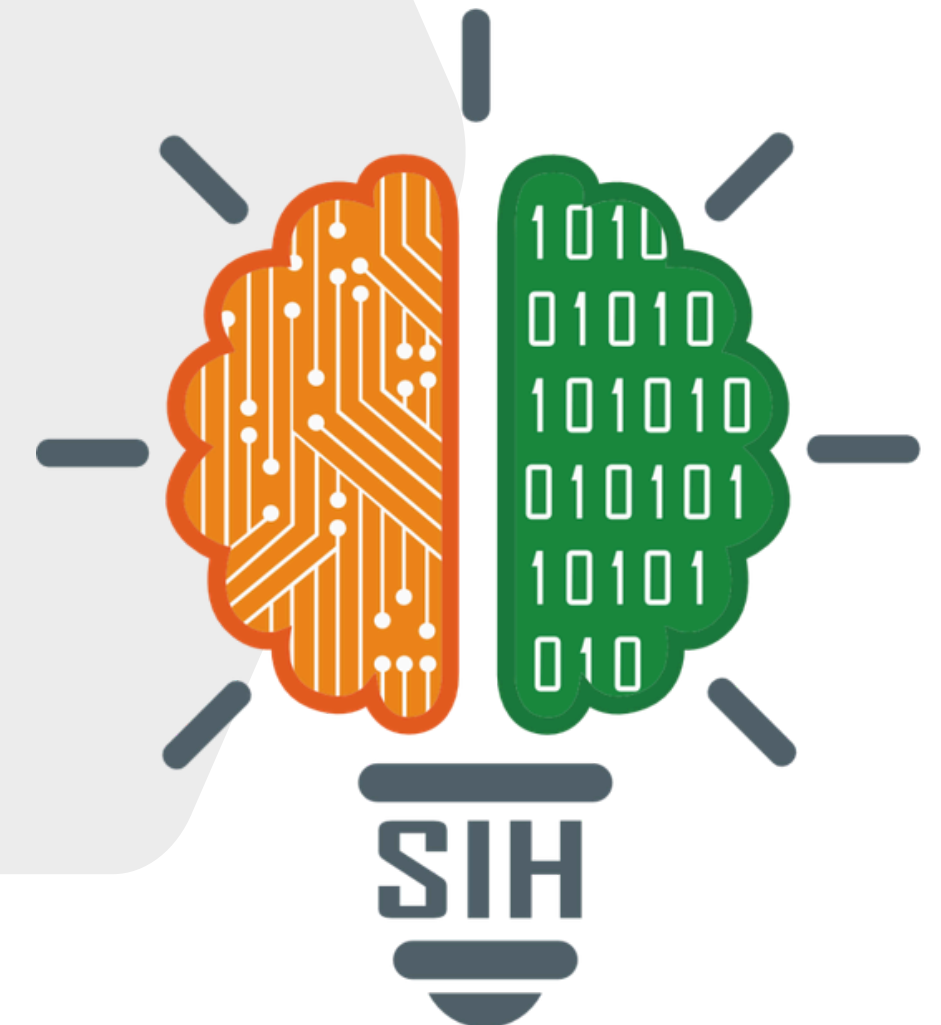


TEAM NAVEDA

- Problem Statement ID – SIH26032
- Problem Statement Title- Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal) :

Naveda - From Ideas to Innovation



! THE PROBLEM

A farmer brings his harvest to the procurement centre, but often finds a long queue, changed schedules, or no clear information about his turn. After hours of waiting, he may have to return without selling his crop.



**FARMERS DON'T KNOW
WHEN TO COME, WHERE
TO GO AND WHAT TO
EXPECT.**

- ✗ Time wastage & long queues
- ✗ Uncertain procurement schedule
- ✗ Lack of real-time information
- ✗ Mental stress and travel cost

*They grow the food,
but still lack the right information.*

💡 THE SOLUTION

AgriSaathi is a simple, farmer-friendly platform that brings the entire procurement journey into one place. Instead of making farmers physically visit centres just to get information, it helps them prepare before they leave, make better decisions while travelling, and track their procurement afterwards.



Real-Time Information

Know your queue, slot & centre status.



Plan Your Visit

Get directions, timing and expected wait time.



Stay Updated

About schedule changes, announcements & alerts.



Track After Sale

Monitor your procurement status and payment.

*More Information → Better Planning
→ Higher Earnings*

🎯 Smarter Way to Sell

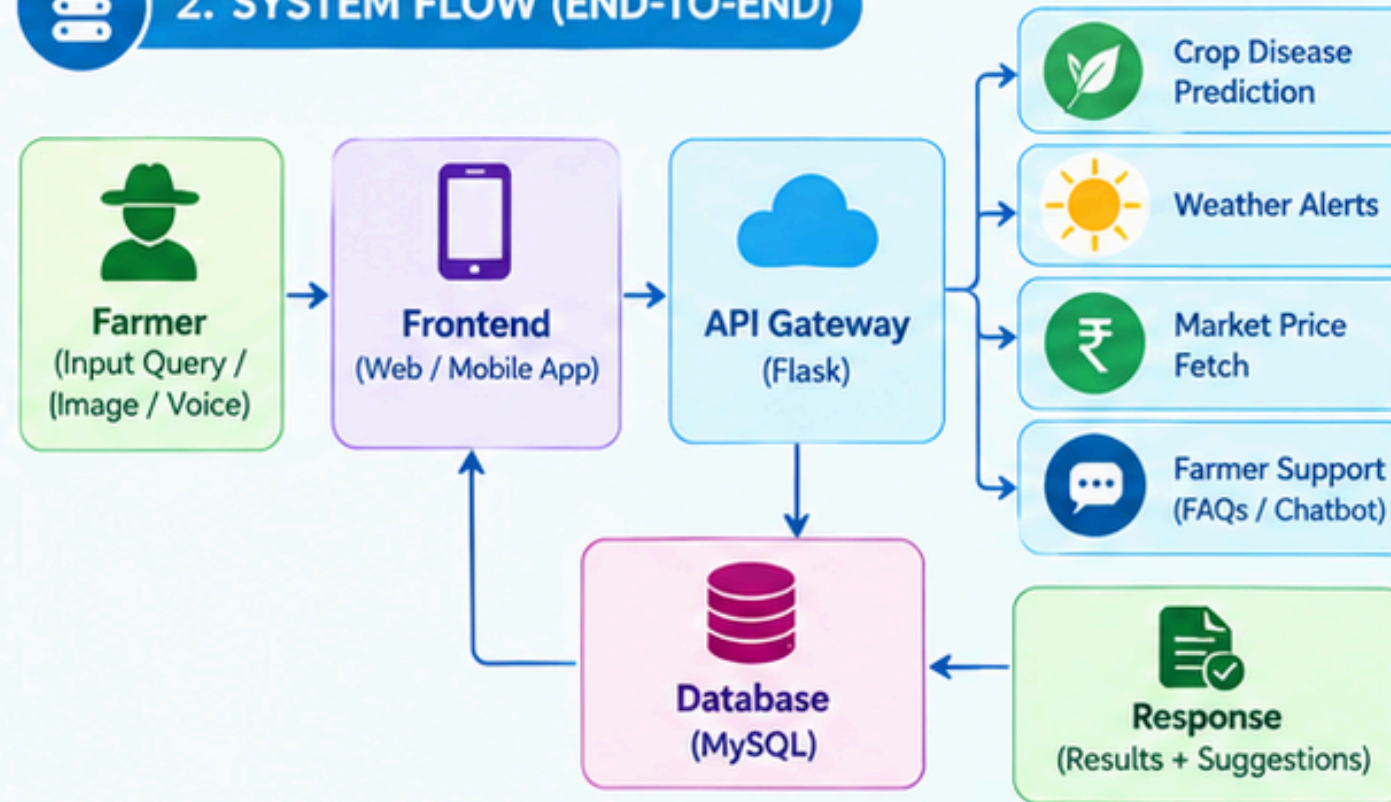
- 1 Virtual Queue**
Check your position in the queue from your phone.
- 2 Best Time to Visit**
Get suggested time slots based on centre crowd & availability.
- 3 Smart Centre Recommendation**
Find the nearest and most suitable procurement centre.
- 4 Real-Time Procurement Tracking**
Track your crop from arrival to payment.
- 5 Complete Farmer Journey**
From registration to payment – all in one place.
- 6 Regional-Language Voice Support**
Get help in your language, anytime.
- 7 Ramu AI Voice Assistant**
Ask questions, get updates, navigate easily.
- 8 Harvest Profit Tracker**
See your estimated earnings and past trends.
- 9 Government Schemes & Subsidies**
Know about available benefits and apply easily.

Digital Support ✦ Stronger Farmers ✦ A Brighter Future

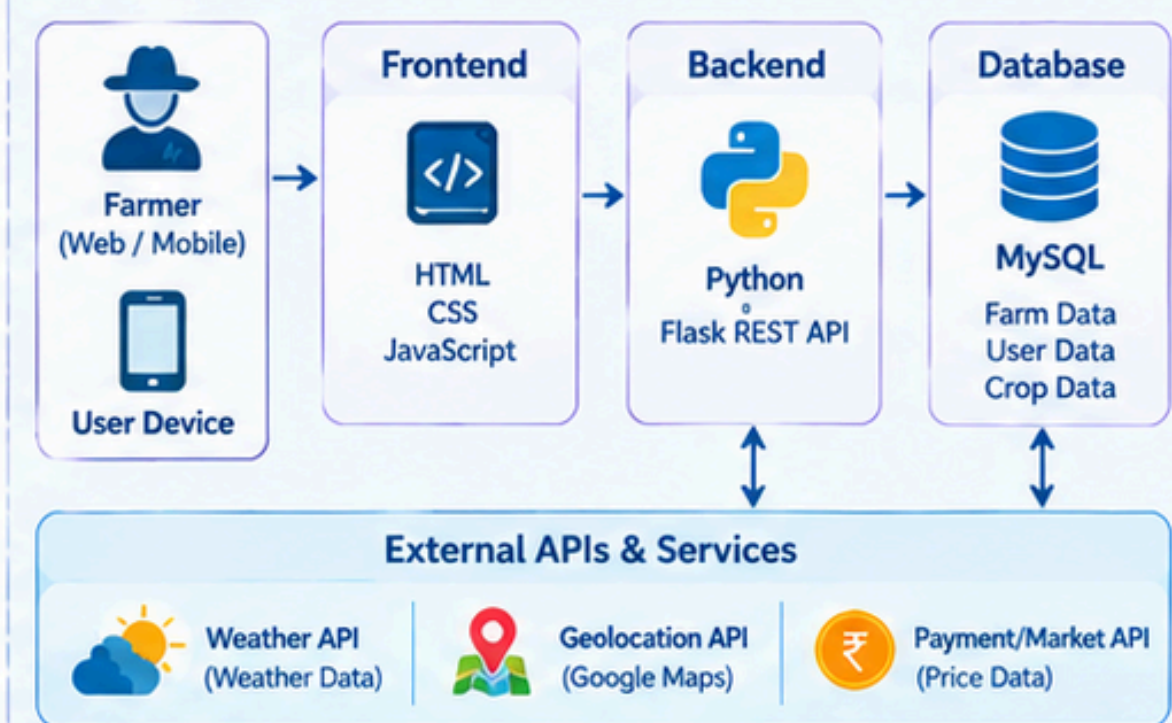
1. USER INTERFACE & FRONTEND



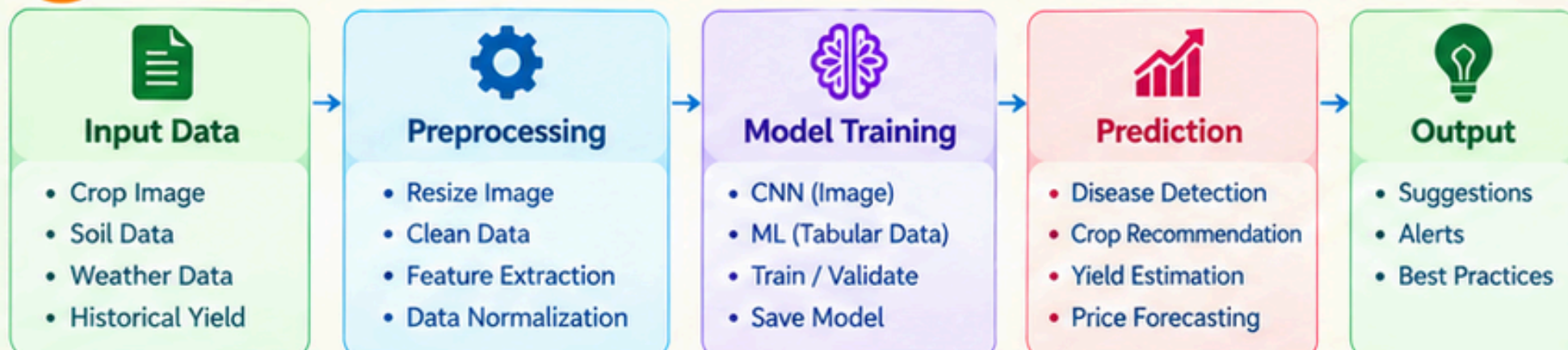
2. SYSTEM FLOW (END-TO-END)



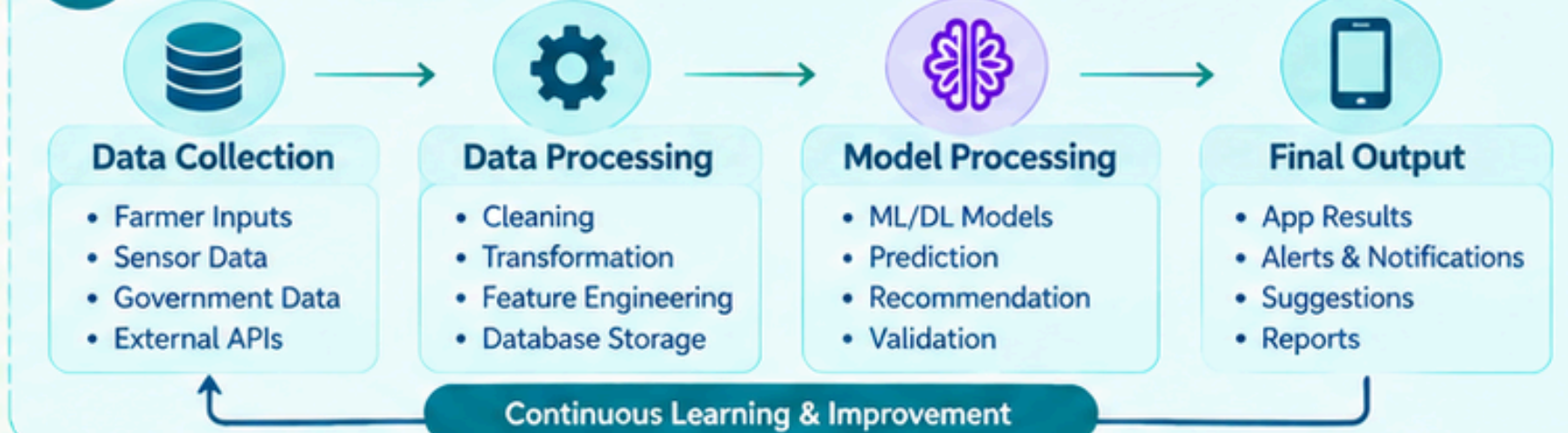
3. ARCHITECTURE DIAGRAM



4. AI/ML MODEL FLOW



5. DATA FLOW



Better Farming



Higher Income



Stronger India

3. Potential Challenges & Risks

Challenge / Risk	Our Solution
 Poor internet connectivity in rural areas	→ Lightweight, low-data platform + SMS fallback
 Digital literacy barriers	→ Simple UI + regional voice support
 Changing procurement schedules	→ Real-time admin updates + notifications
 Inaccurate queue estimates	→ Regular data updates + centre-side monitoring
 Data privacy & security	→ Secure authentication + protected database
 Limited availability of live government data	→ Verified data sources + admin-controlled updates



1. Feasibility — Can AGRISAATHI be built?

	Web Platform Accessible on smartphones and low-end devices
	Frontend HTML, CSS, JavaScript
	Backend Python + Flask
	Database MySQL
	AI/ML Queue prediction & smart recommendations
	Maps & Location Geolocation + Maps API
	Voice Speech-to-Text + Text-to-Speech
	Cloud Deployment Scalable web hosting
	Admin Dashboard Update schedules, slots & status

2. Viability — Can it work in the real world?

	Low-cost Uses existing smartphones and web technology
	Scalable Can expand from one procurement centre to multiple states
	Accessible Regional-language + voice-based interaction
	Practical Reduces unnecessary visits and physical waiting
	Expandable Integrates with government/mandi data sources as APIs

Additional Aspects for Feasibility & Viability

 Analysis of the Feasibility of the Idea - Market demand and user needs - Technical, operational and legal feasibility - Scalability and long-term potential	 Impact Assessment - Social (farmer welfare, rural development) - Environmental (sustainable agriculture) - Economic (increased income, reduced losses)	 Legal & Regulatory Feasibility - Compliance with data privacy laws (e.g., DPDPA Act) - Adherence to government and institutional policies - Licensing and regulatory approvals (if required)
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Reduced Waiting Time

Faster processing at procurement centres.



Fewer Unnecessary Trips

Saves time & travel costs for farmers.



Better Centre & Time Selection

Helps farmers choose the most convenient options.



Greater Transparency

Real-time tracking & visibility in procurement status.



Improved Harvest Planning

Helps farmers plan better and get higher returns.



Improved Access to Digital Services

Brings essential services to rural communities.



Social

Easier and more accessible procurement for farmers.

- ✓ Simplifies access to procurement centres and services.
- ✓ Supports regional-language and voice interaction.
- ✓ Increases inclusion and empowerment of rural farmers.



Economic

Saves time, reduces costs and supports better earnings.

- ✓ Reduces travel and transportation costs.
- ✓ Helps farmers plan sales and get better prices.
- ✓ Supports efficient resource utilization and higher market access.



Digital

Brings technology to rural India, making services simpler.

- ✓ Enables easy access to procurement data and services.
- ✓ Provides real-time updates, schedules and notifications.
- ✓ Builds a user-friendly, platform-driven ecosystem for farmers.



Administrative

Improves coordination, transparency and governance.

- ✓ Better coordination between farmers and procurement centres.
- ✓ Real-time visibility of queue, slots and status.
- ✓ Streamlines data management and reduces manual work.



Research Journey

1



Farmer pain points

- Long waiting times
- Schedule uncertainty
- Unclear procurement status

2



Existing digital agriculture services

- eNAM
- Digital India

3



Queue-management research

- Virtual tokens
- Slot allocation
- Waiting-time prediction

4



Farmer-first design

- Regional language
- Simple UX
- Low digital literacy

5



Solution direction

- Real-time queue/status
- Notifications
- Centre & time selection

Key References



India Meteorological Department (IMD) Agromet Advisory Services

Reference for district-level weather forecasts, agricultural advisories and rainfall information.

https://mausam.imd.gov.in/responsive/agromet_adv_ser_district_current_en.php



e-NAM – National Agriculture Market

Reference for mandi/market information, price discovery and commodity details.

<https://enam.gov.in/>



Agmarknet – Agricultural Market Information

Reference for mandi prices, arrivals and market trends.

<https://agmarknet.gov.in/>



ICAR – Indian Council of Agricultural Research

Reference for crop-related research and agricultural knowledge.

<https://icar.gov.in/>



Digital India

Reference for digital services and rural accessibility.

<https://www.digitalindia.gov.in/>



Research



Problem Identification



Solution Design



Prototype